

1. Introduction

Sentiment analysis is a Natural Language Processing (NLP) task that aims to classify the sentiment expressed in a text, typically as positive or negative. It is widely used for applications such as product review analysis, customer feedback evaluation, and opinion mining.

Different neural network architectures process textual information in different ways. Some models focus on extracting features from the text, while others attempt to capture sequential relationships between words or identify the most important parts of a document.

The objective of this project is to compare four neural network architectures for sentiment classification on the IMDB movie review dataset: a Feedforward Neural Network (FNN), a Recurrent Neural Network (RNN), a Long Short-Term Memory Network (LSTM), and an Attention-based model. All models were trained using the same preprocessing pipeline and experimental settings to ensure a fair comparison.

Performance was evaluated using accuracy, precision, recall, and F1-score. The goal was to investigate how different architectural approaches affect sentiment classification performance on movie reviews.

2. Background

2.1 Feedforward Neural Networks (FNN)

Feedforward Neural Networks (FNNs) are among the simplest neural network architectures. Information flows from the input layer to the output layer without recurrent connections. In this project, word embeddings are averaged to obtain a fixed-length representation of a review, which is then classified using a linear layer. Although FNNs do not model word order, they can perform well on sentiment analysis because sentiment is often associated with specific words and phrases.

2.2 Recurrent Neural Networks (RNN)

Recurrent Neural Networks (RNNs) process text sequentially and maintain a hidden state that is updated after each token. This allows information from previous words to influence later predictions. However, standard RNNs often struggle with long sequences due to the vanishing gradient problem, making it difficult to retain information over many time steps.

2.3 Long Short-Term Memory Networks (LSTM)

Long Short-Term Memory Networks (LSTMs) were developed to address limitations of standard RNNs. Through memory cells and gating mechanisms, LSTMs can preserve important information across longer sequences. As a result, they are generally more effective than vanilla RNNs for many natural language processing tasks.

2.4 Attention Mechanisms

Attention mechanisms allow a model to assign different importance weights to different tokens within a sequence. Instead of relying solely on a final hidden representation, attention enables the model to focus on the most informative words when making predictions. This approach has become highly influential in modern NLP systems and is particularly useful for sentiment classification.

3. Dataset and Preprocessing

3.1 IMDB Dataset

The experiments were conducted using the IMDB movie review dataset, which contains 50,000 labeled reviews equally divided between positive and negative sentiment. The dataset provides predefined training and test splits of 25,000 reviews each. To enable model selection, the training set was further divided into a training subset of 20,000 reviews and a validation subset of 5,000 reviews. The official test set remained untouched until final evaluation.

<i>Dataset</i>	<i>Reviews</i>
<i>Training</i>	20,000
<i>Validation</i>	5,000
<i>Test</i>	25,000

3.2 Text Preprocessing

A common preprocessing pipeline was applied to all models. Reviews were converted to lowercase and tokenized using a regular-expression-based tokenizer. A vocabulary was constructed exclusively from the training subset to avoid information leakage. The vocabulary was limited to the 20,000 most frequent tokens, with special <pad> and <unk> tokens used for padding and unknown words.

All reviews were encoded as sequences of token indices. To enable batch processing, sequences were truncated or padded to a fixed length of 200 tokens. A custom PyTorch Dataset class and DataLoader were used to efficiently generate mini-batches during training.

4. Experimental Setup

To ensure a fair comparison, all neural network architectures were trained using the same dataset split, preprocessing pipeline, and optimization settings. The primary difference between experiments was the model architecture itself.

Table 2 summarizes the most important hyperparameters used throughout the experiments.

<i>Parameter</i>	<i>Value</i>
<i>Vocabulary Size</i>	20,000
<i>Embedding Dimension</i>	100
<i>Maximum Sequence Length</i>	200
<i>Batch Size</i>	32
<i>Learning Rate</i>	0.001
<i>Number of Epochs</i>	3
<i>Optimizer</i>	Adam
<i>Loss Function</i>	CrossEntropyLoss

Model performance was evaluated on the validation set after each training epoch and on the official test set after training was completed. Four evaluation metrics were used: accuracy, precision, recall, and F1-score. While accuracy provides an overall measure of classification performance, precision, recall, and F1-score offer additional insight into the behavior of each model and help identify class-specific prediction tendencies.

All experiments were implemented in Python using the PyTorch framework and executed on a CPU-based environment.

5. Results

5.1 Validation Performance

During training, model performance was monitored using the validation set containing 5,000 reviews. Table 1 presents the final validation accuracy obtained after three training epochs.

<i>Model</i>	<i>Validation Accuracy</i>
<i>FNN</i>	84.94%
<i>RNN</i>	48.48%
<i>LSTM</i>	68.66%
<i>Attention</i>	84.54%

The Feedforward Neural Network achieved the highest validation accuracy, closely followed by the Attention-based model. The LSTM performed substantially better than the standard RNN but remained significantly below the performance of the FNN and Attention models. The RNN achieved validation accuracy close to random guessing, indicating difficulties learning useful representations from the review texts.

5.2 Test Performance

Final model evaluation was performed on the official IMDB test set containing 25,000 previously unseen reviews. Table 2 presents the resulting test accuracies.

Model	Test Accuracy
FNN	83.89%
RNN	49.89%
LSTM	68.25%
Attention	83.58%

The ranking observed during validation remained consistent on the test set. The FNN achieved the highest accuracy, while the Attention model produced very similar performance. The LSTM improved considerably over the standard RNN but remained well below the two best-performing architectures.

Figure 1 presents a visual comparison of training, validation, and test accuracy across all models.

5.3 Precision, Recall, and F1 Score

To obtain a more comprehensive evaluation, precision, recall, and F1-score were computed for each model. The results are shown in Table 3.

Model	Precision	Recall	F1 Score
FNN	83.91%	83.86%	83.89%
RNN	49.93%	74.56%	59.81%
LSTM	74.59%	55.36%	63.55%
Attention	81.32%	87.18%	84.15%

The Attention model achieved the highest F1-score despite slightly lower accuracy than the FNN. This indicates a more balanced trade-off between precision and recall.

The FNN exhibited highly balanced performance across all evaluation metrics, suggesting stable classification behavior for both sentiment classes.

The LSTM achieved relatively high precision but lower recall, indicating that it was more conservative when predicting positive reviews. In contrast, the RNN achieved a relatively high recall but very low precision, suggesting that it frequently predicted the positive class and produced many incorrect classifications.

Figure 2 illustrates the comparison of precision, recall, and F1-score for all architectures.

6. Discussion

6.1 Performance of the Feedforward Neural Network

Despite being the simplest architecture, the FNN achieved the highest test accuracy (83.89%). Although it ignores word order, sentiment classification often depends

strongly on sentiment-bearing words. The results suggest that averaging word embeddings is sufficient to capture much of the information required for IMDB sentiment classification.

6.2 Limitations of the Recurrent Neural Network

The RNN achieved only 49.89% test accuracy, which is close to random guessing. This result is likely related to the vanishing gradient problem, which makes it difficult for standard RNNs to preserve information across long movie reviews. The model also showed an imbalance between precision and recall, indicating unstable classification behavior.

6.3 Improvements Introduced by LSTM

The LSTM achieved a test accuracy of 68.25%, substantially outperforming the standard RNN.

This improvement is consistent with the design objectives of LSTM networks. By introducing memory cells and gating mechanisms, LSTMs are able to preserve relevant information over longer sequences and reduce the impact of vanishing gradients.

Although the LSTM did not reach the performance level of the FNN or Attention model, the results demonstrate that explicit memory mechanisms significantly improve sequence modeling compared to a standard recurrent architecture. The improvement from approximately 50% accuracy to over 68% accuracy highlights the practical value of gated recurrent networks for natural language processing tasks.

6.4 Effectiveness of the Attention Mechanism

The Attention-based model achieved a test accuracy of 83.58% and the highest F1-score of 84.15%.

Unlike recurrent architectures that compress information into a single hidden representation, the attention mechanism allows the model to assign different importance weights to individual tokens. This enables the classifier to focus on sentiment-bearing words while reducing the influence of less informative tokens.

The high recall achieved by the Attention model indicates that it was particularly effective at identifying positive reviews. At the same time, the model maintained strong precision, resulting in the best overall F1-score among all evaluated architectures.

These findings support the idea that attention mechanisms provide an efficient way of identifying relevant information within a text without relying on sequential processing. For sentiment analysis, where a relatively small number of words often determine the overall sentiment of a review, attention appears particularly well suited.

6.5 Overall Comparison

The final ranking of the evaluated architectures was:

1. FNN (83.89% accuracy)
2. Attention (83.58% accuracy)
3. LSTM (68.25% accuracy)
4. RNN (49.89% accuracy)

The results indicate that model complexity alone does not guarantee superior performance. While the recurrent architectures were theoretically capable of modeling sequential information, they did not outperform the simpler alternatives under the experimental conditions used in this project.

The strongest results were obtained by architectures that either aggregated information efficiently (FNN) or explicitly learned which tokens were most important for classification (Attention). This suggests that sentiment analysis on the IMDB dataset relies heavily on identifying informative words and phrases rather than modeling long-range sequential dependencies.

7. Conclusion

This project compared four neural network architectures for sentiment classification on the IMDB movie review dataset: a Feedforward Neural Network (FNN), a Recurrent Neural Network (RNN), a Long Short-Term Memory Network (LSTM), and an Attention-based model. All architectures were trained and evaluated using the same preprocessing pipeline, vocabulary construction method, dataset split, and optimization settings to ensure a fair comparison.

The experimental results demonstrated substantial differences in performance between the evaluated models. The Feedforward Neural Network achieved the highest test accuracy of 83.89%, while the Attention-based model achieved a very similar accuracy of 83.58% and the highest F1-score of 84.15%. The LSTM significantly outperformed the standard RNN, achieving a test accuracy of 68.25% compared to 49.89% for the vanilla recurrent architecture.

These findings indicate that simple approaches can remain highly effective for sentiment analysis tasks. The strong performance of the FNN suggests that the presence of sentiment-bearing words is often sufficient for accurate classification, even when word order is not explicitly modeled. The Attention model further demonstrated the benefits of focusing on the most informative tokens within a review, leading to the best balance between precision and recall.

In contrast, the poor performance of the standard RNN highlights the challenges associated with learning long-range dependencies in lengthy text sequences. Although the LSTM improved substantially upon the RNN through its memory and gating

mechanisms, it was unable to match the performance of the FNN and Attention architectures under the experimental conditions used in this study.

Overall, the results suggest that effective identification and weighting of important sentiment-related words can be more valuable for movie review classification than complex sequential processing. Future work could investigate larger datasets, pre-trained word embeddings, transformer-based architectures, or additional attention mechanisms to further improve performance.

The project successfully demonstrated how different neural network architectures influence sentiment classification performance and provided practical insight into the strengths and limitations of several important deep learning approaches for natural lang language processing.

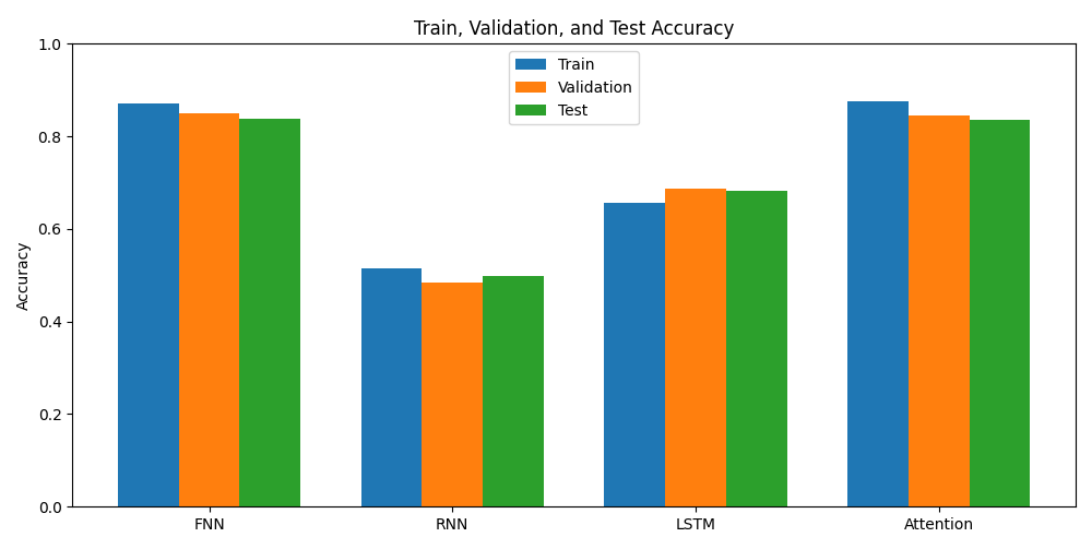


Figure 1



Figure 2

